

ECE 461 - HW 04

1.a) $\underbrace{b_1 \oplus b_2 \oplus b_3}_{k=3} \oplus c_1 = 0 \Rightarrow 1 \text{ parity bit is added to make the codeword even parity.}$

Then $c_1 = b_1 \oplus b_2 \oplus b_3$

	b_1	b_2	b_3	c_1
→	0	0	0	0
→	0	0	1	1
→	0	1	0	1
→	0	1	1	0
→	1	0	0	1
→	1	0	1	0
→	1	1	0	0
→	1	1	1	1

The list of all $2^3 = 8$ possible codewords in the $(4,3)$ codebook

1.b) Single parity check code only ensures codeword has even parity

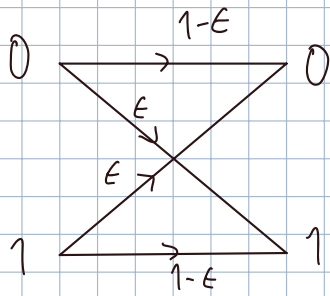
→ Receiver checks the parity by summing (modulo 2) all bits.

↳ If there are even number of errors ($\Rightarrow 2$ or 4 bits flipped), received code will still have even parity. Hence, receiver will incorrectly think that message is valid.

↳ If there are odd number of errors ($\Rightarrow 1$ or 3 bits flipped), on the other hand, received code will have odd parity. Receiver will detect the error.

$(4,3)$ code detect only if there are odd number of errors.

1.c)



★ We have seen in 2.b that an error will go undetected if there is an even number bit flips

→ Then we consider the cases where there are 2 and 4 bit flips.

$$P(\text{undetected}) = P(2 \text{ bit flips}) + P(4 \text{ bit flips})$$

* Having n bits flipped in a sequence of N bits follows Binomial dist.

$$n \sim \text{Binomial}(\epsilon, N)$$

$$P(\text{undetected}) = \binom{4}{2} \epsilon^2 (1-\epsilon)^2 + \binom{4}{4} \epsilon^4 (1-\epsilon)^0$$

$$= 6 \times 0.0001 \times (0.99)^2 + 10^{-8} \approx$$

$$5.88 \times 10^{-4}$$

2)
 $(6,3)$ systematic linear block code formed using $\begin{cases} C_4 = b_1 \oplus b_2 \\ C_5 = b_1 \oplus b_2 \oplus b_3 \\ C_6 = b_1 \oplus b_3 \end{cases}$

Let $\vec{b} = [b_1 \ b_2 \ b_3]^T$ and $\vec{c} = [b_1 \ b_2 \ b_3 \ c_1 \ c_2 \ c_3]^T$ with $G\vec{b} = \vec{c}$

We should then have $G = \begin{bmatrix} I_{3 \times 3} \\ P_{3 \times 3} \end{bmatrix}$ where I is identity and rows of P formed as

$$P_{ij} = \begin{cases} 1, & \text{if } b_j \text{ is in modula sum of } C_{i+3} \\ 0, & \text{else} \end{cases}$$

For example first row of P is $[1 \ 1 \ 0]$ since

$$C_4 = [1 \ 1 \ 0] \vec{b} = b_1 \oplus b_2$$

Following the same logic for other rows,

$$G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

2.b) To construct all possible code words we write all possible 3-bit combinations and append c_4, c_5, c_6 according to their defn.

	b_1	b_2	b_3	$b_1 \oplus b_2$	$b_1 \oplus b_2 \oplus b_3$	$b_1 \oplus b_3$
→	0	0	0	0	0	0
→	0	0	1	0	1	1
→	0	1	0	1	1	0
→	0	1	1	1	0	1
→	1	0	0	1	1	1
→	1	0	1	1	0	0
→	1	1	0	0	0	1
→	1	1	1	0	1	0

All 8 possible code words in the (6,3) codebook

* Notice that minimum Hamming Distance $d_{\min} = 3$

2.c) Let us first construct parity check matrix $\vec{H}_{3 \times 6}$ s.t.

$$H G = 0$$

$$G = \begin{bmatrix} I_{3 \times 3} \\ P_{3 \times 3} \end{bmatrix} \Rightarrow H = [P_{3 \times 3} \quad I_{3 \times 3}] = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Decoded code is then found as $\vec{y} = H \vec{c}$

$$\vec{y} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Decoded code word matches with the 6th column of H , meaning that error is in the 6th bit c_6

* 6th bit c_6 is erroneous and correct codeword (transmitted message) was

$$\hat{y} = 010110$$

Alternative solution

For $\vec{y} = 010111$, we can check the parity bits

$$c_4 = b_1 \oplus b_2 = 0 \oplus 1 = 1 \quad \checkmark \Rightarrow c_4 = 1 \text{ indeed, meaning } b_1 \text{ and } b_2 \text{ are correct.}$$

$$c_5 = b_1 \oplus b_2 \oplus b_3 = 0 \oplus 1 \oplus 0 = 1 \quad \checkmark \Rightarrow c_5 = 1 \text{ indeed, meaning } b_3 \text{ is correct as well}$$

$$c_6 = b_1 \oplus b_3 = 0 \oplus 0 = 0 \neq 1 \quad \times \Rightarrow \text{Since, we've found that neither } b_1 \text{ nor } b_3 \text{ is erroneous } c_6 \text{ itself is incorrect,}$$

* Note that we made the assumption that only 1 bit of error exists. (at least we can correct 1 bit error) since $d_{\min} = 3 = 2l + 1 \Rightarrow l = 1$.

3)

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} P \\ I_4 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} I_3 & P \end{bmatrix}$$

3.a)

$$G^T H^T = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \boxed{0_{4 \times 3}} \Rightarrow \text{all zeros matrix}$$

Easier way to see it would be:

$$G^T H^T = (HG)^T = \left[\begin{bmatrix} I_3 & P \end{bmatrix} \begin{bmatrix} P \\ I_4 \end{bmatrix} \right]^T = \left(\underbrace{I_3 P}_P + \underbrace{P I_4}_P \right)^T = (2P)^T = (0_{3 \times 4})^T = 0_{4 \times 3}$$

* Notice that $(P+P)_{ij} = 0 \quad \forall i, j$ since $1 \oplus 1 = 0 \oplus 0 = 0$ in modulo 2 arithmetics.

3.b) Dual code will be $(7, 3)$ codewords with generator matrix

	b_1	b_2	b_3	$b_1 \oplus b_2$	$b_2 \oplus b_3$	$b_1 \oplus b_2 \oplus b_3$	$b_1 \oplus b_3$
0-	0	0	0	0	0	0	0
1-	0	0	1	0	1	1	1
2-	0	1	0	1	1	1	0
3-	0	1	1	1	0	0	1
4-	1	0	0	1	0	1	1
5-	1	0	1	1	1	0	0
6-	1	1	0	0	1	0	1
7-	1	1	1	0	0	1	0

$$H^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{array}{l} \rightarrow b_1 \\ \rightarrow b_2 \\ \rightarrow b_3 \\ \rightarrow c_4 = b_1 \oplus b_2 \\ \rightarrow c_5 = b_2 \oplus b_3 \\ \rightarrow c_6 = b_1 \oplus b_2 \oplus b_3 \\ \rightarrow c_7 = b_1 \oplus b_3 \end{array}$$

8 codewords
in the dual code

$$d_{\min} = \min_{r, k, r \neq k} d(c_r, c_k)$$

3.c) Consider any 2 words whose pure bits (first 3 bits) have a Hamming distance of 1. For example

$$\begin{aligned} \hookrightarrow d(0000000, 0010111) &= 4 \\ \hookrightarrow d(1100101, 1110010) &= 4 \end{aligned} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} 1 \text{ data bit difference} \\ + \\ 3 \text{ parity bit difference} \end{array}$$

• Now, consider the case 2 data bits are different

$$\hookrightarrow d(0010111, 0101110) = 4 \quad \left. \right\} 2 \text{ data} + 2 \text{ parity bit difference}$$

• In the case all 3 data bits are different

$$\hookrightarrow d(0000000, 1110010) = 4 \quad \left. \right\} 3 \text{ data} + 1 \text{ parity bit diff.}$$

$$d_{\min} = 4$$